

# STATS 507 final project

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## 1 Overview

Image classification is a popular topic in computer vision and is widely used in various fields such as medical diagnosis, wildlife monitoring, e-commerce and autonomous driving. I am interested in this field, so I am going to find a suitable dataset to classify using Python. For this project, I chose the dataset provided by Hugging Face: mertcobanov/animals dataset, a multi-class dataset containing labeled animal images. Firstly, the image data is collected, and I performed data analysis on the dataset. I am also interested in data augmentation, and I am going to use some computer vision data augmentation methods, such as basic data augmentation methods like rotation, flipping, cropping, etc; Advanced data augmentation methods such as Cutmix, Mixup and AutoAugment. Besides, I will try to use various classification methods, including traditional machine learning methods such as Support Vector Machines (SVM) and Convolutional Neural Networks (CNN), and try to use the pre-trained model provided by Hugging Face. It should be noted that I mainly use traditional machine learning methods due to the lack of local CPU arithmetic, other methods require more time for local experimentation and the local machine configuration is not enough. The flow of this project is shown in Figure 1.

Through this project, I want to understand model performance and optimization: Understand the advantages and disadvantages of different algorithms (e.g., CNN, SVM) and applicable scenarios. Explore the effectiveness of model optimization methods (e.g., data augmentation, feature extraction) in improving classification accuracy. Try to use deep learning frameworks (e.g., PyTorch or TensorFlow) to implement model training and apply it to image classification. Observe the effect of data quality (e.g., image clarity, category balance) on the classification effect.

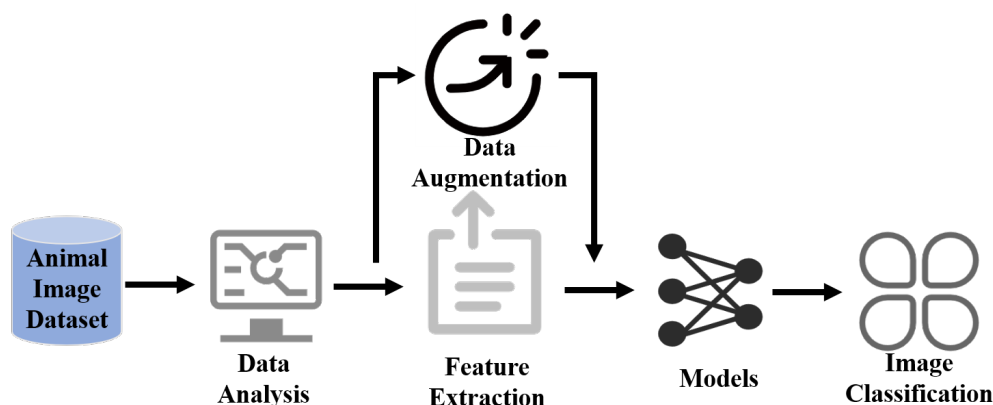


Figure 1: Project Flowchart

## 2 Prior Work

### 2.1 Literature Review

Traditional Machine Learning for Image Classification: Support Vector Machines (SVM) have been widely used for image classification tasks before the deep learning era. They are particularly effective in cases where the dataset is small but features are discriminative. Studies have shown that combining SVMs with robust feature extraction methods, such as using pre-trained convolutional neural networks (CNN) like ResNet, can achieve competitive performance in scenarios with limited data. (Niu et al., 2012) proposed a hybrid model that integrates the synergy of two advanced classifiers: convolutional neural network (CNN) and support vector machine (SVM). In this model, CNN acts as a trainable feature extractor and SVM acts as a recognizer. This hybrid model automatically extracts features from the original image and generates predictions. Experiments were conducted on the well-known MNIST digital database. Comparison with other studies shows that this fusion achieves better results[1]. (Ahlawat et al., 2020) proposed a hybrid CNN-SVM model where CNN is used as an automatic feature extractor and SVM is used as a binary classifier. The sensory field of CNN helps to automatically extract the most distinguishing features from these handwritten digits[2]. Automatic classification of electrocardiogram (ECG) signals is crucial for diagnosis and treatment. (Ozaltin et al., 2022) proposes a novel CNN to automatically extract features from images. and combines it with SVM, the classification accuracy is greatly improved. The framework can help clinicians to perform ECG type recognition[3].

Convolutional Neural Networks (CNN): CNN revolutionized image classification with architectures like LeNet, AlexNet, and ResNet. The introduction of ResNet (He et al., 2016) addressed the problem of vanishing gradients in deep networks by introducing residual connections[4]. ResNet50 is a powerful image classification model that can be trained on large datasets and achieve state-of-the-art results.

Data augmentation can enhance the training of the model, the effect diagram is shown in Figure 2. Data augmentation techniques like Mixup (Zhang et al., 2018), it generates new training samples by mixing two images and their corresponding labels in a certain ratio[5]. CutMix (Yun et al., 2019), it generates new training samples by cutting and pasting part of one image into another while scaling the labels[6]. AutoAugment (Cubuk et al., 2019) automatically learns an optimal set of data augmentation strategies. It aims to augment the diversity of the data by searching for combinations of augmentation strategies to find the augmentation method that maximizes the performance of the validation set[7]. These methods have been shown to improve generalization and robustness by generating synthetic training samples. These methods help mitigate overfitting and improve model performance, especially in datasets with limited diversity.

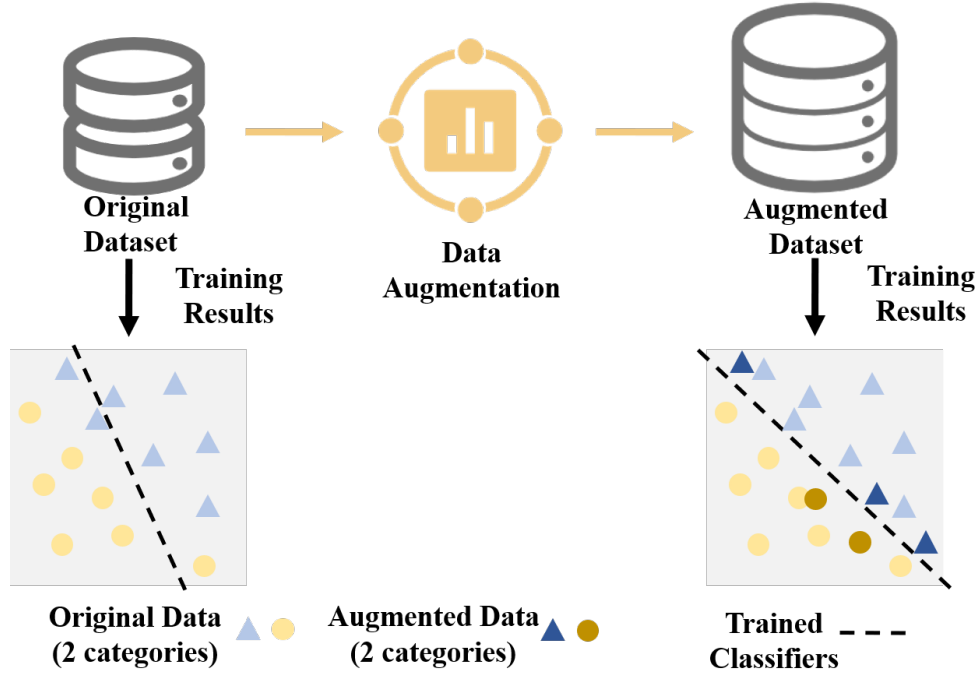


Figure 2: Data Augmentation Effect Diagram

## 2.2 Proposed Methods

- **Dataset Processing and Analysis**

Load the dataset (mertcobanov/animals) and inspect the number of classes, distribution of images across classes, and image dimensions. Identify potential issues such as class imbalance, low-quality images, or inconsistencies in image resolutions. Resize all images to a fixed resolution (e.g., 224x224) to ensure compatibility with pre-trained models. Normalize pixel values to standardize the input for deep learning models. Besides, visualize samples from each class to understand dataset diversity.

- **Data Augmentation**

To improve model robustness and generalization, apply data augmentation techniques. **Basic Augmentation:** Apply transformations such as random rotations, flips, color jittering, and cropping to increase dataset diversity. **Advanced data Augmentation:** Cutmix, Mixup or Autoaugment.

- **Model-Based Classification**

Use different kinds of classification techniques to evaluate performance comprehensively: **Traditional Machine Learning(e.g., SVM):** Train an SVM classifier on the extracted ResNet-18 features. Evaluate its performance in handling high-dimensional features. **Convolutional Neural Networks (CNN):** CNN is one of the main models in image classification tasks. It is capable of automatically extracting meaningful features from an image and using these features for classification. **Hugging Face’s model:** Use Hugging Face’s model(e.g., microsoft/resnet-50) on the dataset. Besides, I can evaluate the impact of models combined with advanced data augmentation.

### 3 Preliminary Results

- **Data understanding**

The mertcobanov/animals dataset contains a total of 5400 training samples. Each animal image is in RGB mode with varying dimensions and the dataset has images of 90 animal species, such as 'antelope', 'cat', 'dog', etc. Each class is perfectly balanced, with 60 images per class, indicating no class imbalance issues. This dataset is of high quality, no imbalance among classes ensures fair representation of all categories in training and evaluation, this dataset contains a wide variety of animal classes, making it suitable for animal classification tasks. But there are potential challenges here. The size of the images varies and some of them are too large, and the uneven image sizes may affect the model performance if they are not preprocessed. The quality and lighting conditions of some images are too poor, which may have an impact on the training of the model. Besides, some of the animals are too similar (e.g., 'squid' vs. 'octopus', 'dog' vs. 'wolf'), which could increase classification complexity. Examples of classification errors are shown in Figure 3.



Figure 3: octopus(52) is misclassified as squid(78) on the right

- **Basic Model**

Directly using SVM to classify images may not be highly accurate, and it's very time-consuming. I need to use principal component analysis (PCA) or other methods to achieve dimensionality reduction. However, this can be solved by using feature extraction. Feature Extraction and SVM use ResNet-18 to extract image features and SVM acts as a recognizer. This is not only efficient and time-saving, but also highly accurate. Training a CNN model directly, this requires a high-performance GPU for efficient training. CNN using this animal dataset takes a long time to train on the local CPU, about 15 minutes for an epoch, and I can only perform a small number of epochs locally for the classification task using this model, which results in the possibility of low classification accuracy. Using Hugging Face's model, Many of these models are CNNs, so it's possible to run into the same problems. Training on these models requires longer epochs, making experiment slower.

- **Tools Used**

NumPy, SciPy, Pyplot, Pandas, Seaborn: Data Analysis and Visualization.

PyTorch: For building and training CNN models and feature extraction using ResNet-18.

Scikit-learn: Provide machine learning algorithms and scikit-learn for accuracy, precision, recall, F1-score, and confusion matrix generation.

Torchvision: Transforms are used for preprocessing and data augmentation. Torchvision offers pre-trained models for various tasks, including image classification. Utilities for image operations, such as saving, reading, and visualizing images.

- **Tools to Explore**

Data Augmentation: Cutmix, Mixup or Autoaugment.

Hugging Face: datasets and models.

## 4 Project Deliverables

- **What will a successful project produce**

A complete workflow for image classification combining different kind of methods: PCA+SVM classification, CNN Feature extraction + SVM classification, CNN, Hugging Face's model(e.g., microsoft/resnet-50). A preprocessed and augmented version of the dataset using basic augmentations or advanced augmentations. A detailed comparison of the performance of SVM, CNN+SVM, CNN, and transformer-based models. Metrics such as accuracy, precision, recall, F1-score, training time and confusion matrices for each model. A well-documented repository, Python scripts for classification tasks. A concise report summarizing the methodology, results, and major gains.

- **What are my sub-goals**

Learn to improve data quality, Provide diverse and clearer inputs to the model. Learn to select appropriate models to improve the accuracy and generalization of classification models. Time and efficiency optimization with rational use of local resources while maintaining model accuracy. Analyze the performance of the classification model for each animal category image and analyze the reasons for misclassification.

## 5 Timeline

Week 1-2: Check out the information, read the literature, and determine the project idea. Data preparation and data analysis.

Week 3-4: Project-specific realization, data augmentation, model training and evaluation.

Week 5: Improvement of project code, perform a detailed analysis of model performance, and prepare the project report.

## References

- [1] Niu, Xiao-Xiao, and Ching Y. Suen. "A novel hybrid CNN-SVM classifier for recognizing handwritten digits." Pattern Recognition 45.4 (2012): 1318-1325.

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